A RESOLUTION OF A NORMAL TWO DIMENSIONAL LOCAL RING YIELDS A RESOLUTION BY NORMALIZED BLOWUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the complete case through the transfer of normality and regularity along completion and the descent of sequences of normalized blowups, the domination of the given resolution first by normalized blowups and then by blowups in closed points yielding finitely many singular points, the propagation of rationality followed by the passage to Gorenstein local rings, and the final resolution of rational double points.

Editorial note. The label lemma-existence-implies-existence-by-normalized-blowing-ups names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-existence-implies-existence-by-normalized-blowing-ups –start 3949 –end 4048 –local-id 28073 returned RESULT=OK (L28073 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag 0BGM, not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Normality and regularity pass to and from the completion. Same chapter, the lemma that the completion of a normal local ring of a regular alteration is normal, and the lemma transferring regularity of a resolution along the completion. Supplies the reduction to the complete case and the transfer of regularity between a sequence and its completion.
- Descending a sequence of normalized blowups from the completion. Same chapter, the lemma that a sequence of normalized blowups of the completion comes from a sequence over the original ring. Supplies the passage from sequences over the completion back to sequences over the ring.
- Domination of a resolution by normalized blowups at closed points. Same chapter, the lemma dominating a proper birational morphism of a two dimensional Noetherian Nagata scheme by normalized blowups at closed points, and the lemma dominating a modification by blowups in points. Supply the sequence of normalized blowups dominating the given resolution and the further sequence of point blowups dominating it.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Propagation of rational singularities and the resolution of rational double points. Same chapter, the lemma that regularity defines a rational singularity, the lemma propagating the rational singularity property, the lemma producing finitely many blowups making the local rings Gorenstein, and the lemma resolving rational double points. Supply the intermediate surface with Gorenstein local rings and rational double point singularities, and its resolution.

Source reference. Original proposition 0335: sources/stacks-chow/tags.txt, source lines 4044–4044. Prerequisite pointer invoked at proof line(s) 3972; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BFR: sources/stacks-chow/tags.txt, source lines 14394–14394. Prerequisite pointer invoked at proof line(s) 3976; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AHI: sources/stacks-resolve/pinned-original-source.tex, source lines 1146–1146. Prerequisite pointer invoked at proof line(s) 3992; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BBT: sources/stacks-resolve/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 3989; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BFY: sources/stacks-resolve/pinned-original-source.tex, source lines 1495–1495. Prerequisite pointer invoked at proof line(s) 4026; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BGA: sources/stacks-resolve/pinned-original-source.tex, source lines 3236–3236. Prerequisite pointer invoked at proof line(s) 3962; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BG6: sources/stacks-resolve/pinned-original-source.tex, source lines 2997–2997. Prerequisite pointer invoked at proof line(s) 3960; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BG0: sources/stacks-resolve/pinned-original-source.tex, source lines 2035–2035. Prerequisite pointer invoked at proof line(s) 4016; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BBV: sources/stacks-resolve/pinned-original-source.tex, source lines 2688–2688. Prerequisite pointer invoked at proof line(s) 4019; it names a standard fact of the same chapter that the counted argument

uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BGI: sources/stacks-resolve/pinned-original-source.tex, source lines 3827–3827. Prerequisite pointer invoked at proof line(s) 3958; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B4Q: sources/stacks-resolve/pinned-original-source.tex, source lines 2205–2205. Prerequisite pointer invoked at proof line(s) 4015; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BGE: sources/stacks-resolve/pinned-original-source.tex, source lines 3716–3716. Prerequisite pointer invoked at proof line(s) 4046; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BFK: sources/stacks-chow/tags.txt, source lines 14388–14388. Prerequisite pointer invoked at proof line(s) 3980; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07QW: sources/stacks-chow/tags.txt, source lines 9846–9846. Prerequisite pointer invoked at proof line(s) 3974; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BFP: sources/stacks-chow/tags.txt, source lines 14392–14392. Prerequisite pointer invoked at proof line(s) 4010; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Assume A is normal and has dimension 2. If* $\mathrm{Spec}(A)$ *has a resolution of singularities, then* $\mathrm{Spec}(A)$ *has a resolution by normalized blowups.*

Proof. By Lemma 0BGI the completion $A^\wedge$ of A is normal. By Lemma 0BG6 we see that $\mathrm{Spec}(A^\wedge)$ has a resolution. By Lemma 0BGA any sequence $Y_n \to Y_{n-1} \to \ldots \to \mathrm{Spec}(A^\wedge)$ of normalized blowups of comes from a sequence of normalized blowups $X_n \to \ldots \to \mathrm{Spec}(A)$. Moreover if Y_n is regular, then X_n is regular by Lemma 0BG6. Thus it suffices to prove the lemma in case A is complete.

Assume in addition A is a complete. We will use that A is Nagata (Algebra, Proposition 0335), excellent (More on Algebra, Proposition 07QW), and has a dualizing complex (Dualizing Complexes, Lemma 0BFR). Moreover, the same is true for any ring essentially of finite type over A. If B is a excellent local normal domain, then the completion $B^\wedge$ is normal (as $B \to B^\wedge$ is regular and More on Algebra, Lemma 0BFK applies). We will use this without further mention in the rest of the proof.

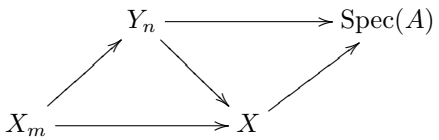
Let $X \to \mathrm{Spec}(A)$ be a resolution of singularities. Choose a sequence of normalized blowing ups

$$Y_n \to Y_{n-1} \to \ldots \to Y_1 \to \mathrm{Spec}(A)$$

dominating X (Lemma 0BBT). The morphism $Y_n \to X$ is an isomorphism away from finitely many points of X. Hence we can apply Lemma 0AHI to find a sequence of blowing ups

$$X_m \to X_{m-1} \to \ldots \to X$$

in closed points such that X_m dominates Y_n. Diagram



To prove the lemma it suffices to show that a finite number of normalized blowups of Y_n produce a regular scheme. By our diagram above we see that Y_n has a resolution (namely X_m). As Y_n is a normal surface this implies that Y_n has at most finitely many singularities $y_1, \ldots, y_t$ (because $X_m \to Y_n$ is an isomorphism away from the fibres of dimension 1, see Varieties, Lemma 0BFP).

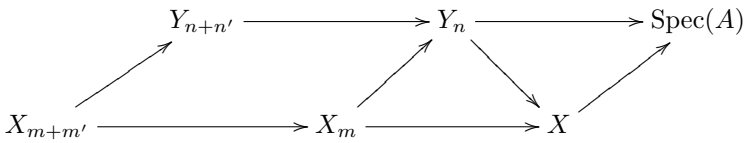
Let $x_a \in X$ be the image of y_a. Then $\mathcal{O}_{X,x_a}$ is regular and hence defines a rational singularity (Lemma 0B4Q). Apply Lemma 0BG0 to $\mathcal{O}_{X,x_a} \to \mathcal{O}_{Y_n,y_a}$ to see that $\mathcal{O}_{Y_n,y_a}$ defines a rational singularity. By Lemma 0BBV there exists a finite sequence of blowups in singular closed points

$$Y_{a,n_a} \to Y_{a,n_a-1} \to \ldots \to \mathrm{Spec}(\mathcal{O}_{Y_n,y_a})$$

such that Y_{a,n_a} is Gorenstein, i.e., has an invertible dualizing module. By (the essentially trivial) Lemma 0BFY with $n' = \sum n_a$ these sequences correspond to a sequence of blowups

$$Y_{n+n'} \to Y_{n+n'-1} \to \ldots \to Y_n$$

such that $Y_{n+n'}$ is normal and the local rings of $Y_{n+n'}$ are Gorenstein. Using the references given above we can dominate $Y_{n+n'}$ by a sequence of blowups $X_{m+m'} \to \ldots \to X_m$ dominating $Y_{n+n'}$ as in the following



Thus again $Y_{n+n'}$ has a finite number of singular points $y'_1, \ldots, y'_s$, but this time the singularities are rational double points, more precisely, the local rings $\mathcal{O}_{Y_{n+n'},y'_b}$ are as in Lemma 0BGE. Arguing exactly as above we conclude that the lemma is true. $\square$